

The Action Space, Not the Decomposition: Why an Interpretable Mixture-of-Drives Over-Brakes—and What Makes It Drive

Dana Kim

Abstract

We study X-MoD, an interpretable Mixture-of-Drives policy for autonomous driving: a router that arbitrates over four named drive-experts—*safety*, *legality*, *comfort*, *efficiency*—so that the gating weights are a human-readable account of *why* the policy acts. We report three results that, together, sharpen what “interpretable driving” actually delivers. **First (explanation works):** with safety-event-aware data collection, X-MoD’s safety gate becomes faithfully risk-aligned—the gate-safety-event correlation reaches **0.80**, with the safety gate mean separating **0.65 on safety events vs 0.04 otherwise**, at **1.5 ms** inference. Critically, this alignment is unlocked by *data composition, not architecture*: without safety-positive samples the identical model shows near-zero correlation. **Second (control fails under raw pedals):** when the same interpretable policy is evaluated *closed-loop* on Bench2Drive-220 it does not drive. It collapses to a constant-brake router state, and an extensive recovery campaign—data scale-up (to $\sim 8.5\text{k}$ frames / 43 routes / 9 towns), loss balancing, capacity increase, and a full structural fork (separated mobility/brake heads, hazard-conditioned inputs, from-scratch retraining)—never escapes a **safety/mobility seesaw**: every configuration either over-brakes (route completion 3–5%) or under-brakes (collisions). **Third (the over-brake was the learned brake; what remains is generalization):** the over-brake comes from the model’s *learned brake output*. An action head with no brake (it *cannot* over-brake) paired with a classical control stack—a PID on the predicted target speed, a standard safety shield, and an anti-stall floor—turns the 3–5% stall into driving. Of the five evaluation routes only one (24211) appears in training; on the four *unseen* routes the policy **completes one at 100% with zero collisions** (24330, robust across three traffic seeds) yet reaches only 18–53% on the others. The gap is *scenario-dependent*, not a clean train/test split—the lone trained route still collides, and the cleanest route is unseen—and it is not closed by *quantity* (a $11 \rightarrow 43$ -route scaling probe leaves mean unseen completion flat), so it is a composition/robustness gap rather than simple overfitting. We conclude that the over-brake was the learned brake output, *not* the interpretable decomposition; what remains is *generalization*, which a hybrid policy-plus-control-stack does not by itself provide.

1 Introduction

“Interpretable autonomous driving” usually means that a model’s internals can be *read*—attention maps, modular sub-policies, named experts. It rarely means that the readable structure has been *tested as a closed-loop controller*. These are different properties, and conflating them is how an interpretable model earns trust it has not demonstrated.

We separate the two with a single model. X-MoD is a transparent Mixture-of-Drives: a router weights four drive-experts (safety / legality / comfort / efficiency), and the gating vector is the model’s own explanation of its decision. We ask a deliberately blunt question: **does a transparent value-arbitration policy that demonstrably *explains* also *drive*?**

Our answer is three-part and, we think, more useful than any single verdict:

- **Explanation: yes.** The safety gate measurably tracks real safety events (correlation 0.80), and we show *why*—it is unlocked by safety-event data, not by the architecture.
- **Control under raw pedals: no.** The same policy, evaluated closed-loop on Bench2Drive-220 [5], cannot reconcile safety and mobility; it sits on a seesaw between over-braking and colliding, and stays there across data scale, loss balancing, added capacity, and a structural redesign.

- **Control with the learned brake removed: it drives.** Zeroing the model’s brake output (so it cannot over-brake), with braking from a standard safety shield, breaks the over-brake corner on the identical model and data (route completion 3–5% $\rightarrow$ 46%); adding a classical target-speed controller then drives an *unseen* route to 100% (zero collisions). The over-brake was the learned brake output, not the decomposition.

The contribution is to hold these results in one frame and *locate* the failure precisely. The over-brake seesaw—which survives data scale, capacity, and structural redesign while the model emits a learned brake—is removed by zeroing that brake output (with a standard safety shield providing the actual braking). So the over-brake was the learned brake, not the decomposition. What remains (lateral control and scenario-dependent generalization—some unseen routes drive cleanly, others fail) names what an interpretable arbitrator is still missing to become a controller.

2 Related Work

Interpretable / privileged driving policies. Learning by Cheating [2] and Roach [6] use privileged teachers and modular structure; we share the transparent-arbitration motivation but evaluate *whether the transparency survives as control*. **Mixture-of-Experts policies.** X-MoD is an MoE policy with *named, semantically meaningful* experts, which is what makes its gate an explanation rather than an opaque weight. **Explainability in driving** typically validates that an internal signal *correlates* with a concept; we do that (the safety gate), then ask the rarely-asked follow-up: does the same model *drive* closed-loop? **Closed-loop benchmarking** [5, 4, 3]: open- and closed-loop scores can diverge sharply; our negative result is a concrete instance—strong offline alignment, zero closed-loop competence.

3 Model: X-MoD

X-MoD is a router over four drive-experts (safety, legality, comfort, efficiency) on a shared perceptual backbone (ResNet34 or a tinyCNN). The forward map is $f(\text{image}_{[B,3,128,128]}, \text{ego}_{[B,2]} = [\text{speed}, \text{wp_angle}]) \rightarrow (\text{action}, \text{gate_probs}, \text{experts})$, where **gate_probs** is the router’s arbitration over the four experts and is the object of explanation. The action head performs **direct regression of throttle / brake / steer**—a design choice we return to in Section 6 as the central confound of the negative result.

4 Part I—Explanation Works

Setup. CARLA 0.9.15 [4], synchronized simulation; data collected with rule-based teachers and event-driven labeling; 11,500 samples. We report action quality (action MSE), routing quality (router accuracy), inference latency, and the explainability target: the correlation between the safety gate and labeled safety events (“H3”).

Result. Table 1 compares three training configurations on the full sample.

Table 1: Explanation: routing loss and safety-event data drive gate–risk alignment.

Run	Action MSE	Router Acc	Latency (ms)	Safety Corr
<code>ce_align_all</code>	0.0205	0.7698	1.624	−0.0301
<code>bce_all</code>	0.2908	0.5642	1.571	0.2392
<code>ce_align_safety_all</code>	0.0379	0.7616	1.543	0.8035

For the aligned run, the safety gate cleanly separates by event: mean **0.6541 on safety events vs 0.0367 otherwise** (safety-event rate 0.1004), at **1.54 ms** inference (Fig. 1). The gate is not merely *active*—it is *conditionally* aligned to safety-critical moments.

The decisive variable is data, not architecture. The same architecture yields correlation −0.03 (`ce_align_all`) when the evaluated distribution lacks safety-positive samples, and 0.80 once safety-event



Figure 1: The safety gate activates on safety events (0.65) and stays quiet otherwise (0.04)—a conditional, not constant, alignment (corr 0.80).

collection and re-labeling are forced. H3 is *unlocked by data composition*; no architectural change is responsible. The transferable lesson: transparent routing can be *measured* and *tied to real risk events*—but only if the data exposes those events.

Honesty. The H3 evidence is primarily offline; baseline diversity is limited; a small closed-loop check (seed 10) shows 0 collisions over short routes (~ 152 m) but is not a driving result. Part II is.

5 Part II—Control Fails

From explanation to control. We wrap the identical interpretable policy as a Bench2Drive-220 leaderboard agent and score it closed-loop. It scores **DS 0**: the router collapses out-of-distribution to the legality expert (gate $\rightarrow [0, 1, 0, 0]$), producing a constant brake. Explanation quality does not transfer to control.

A genuine recovery campaign. We did not stop at the collapse. We ran, in order: (i) teacher-data scale-up to **8,456 frames / 43 routes / 9 towns** (PDM-Lite [1] teacher, failed-teacher runs excluded); (ii) loss balancing for mobility; (iii) a capacity increase (wider router/expert MLPs); and (iv) a full **structural fork**—separating a mobility head from a hazard-conditioned brake head, adding front-actor distance and closing speed to the input, and retraining **from scratch** (no warm start) on the 8.4k data. Every candidate was judged *only* by closed-loop Driving Score.

The seesaw. No configuration reconciles safety and mobility. Table 2 traces the tension on one representative route (24211): pushing the policy to move makes it collide; making it safe makes it stop (Fig. 2).

Table 2: Control: the safety/mobility seesaw at route 24211.

Candidate	route %	collisions	mean brake
scale_b_full	3.08	0	0.586
scale_b_full_go3	100	3	0.022
scale_b_cap1_balanced	4.59	0	0.205
struct_sep_fresh	3.08	0	0.350

The structural fork—the strongest intervention—over-brakes on all three gate routes alike (Table 3): it reaches the goal of *not colliding* only by *not driving*.

Mechanism: fed but ignored. The hazard evidence the brake head was supposed to use *is in the input*, yet the policy does not condition on it: within 10 m of a front actor the mean throttle is **0.93** and the mean brake **0.00**. The policy learns a *global* brake bias—turn it up and it over-brakes everywhere, turn it down and it collides—rather than a *conditional* response to the actual hazard. Two extremes are reachable; the middle is not.

Table 3: Structural fork (separated heads + hazard inputs + from-scratch): over-brake on all routes.

Route	route %	collisions
24211	3.08	0
24240	4.05	0
24330	4.05	0

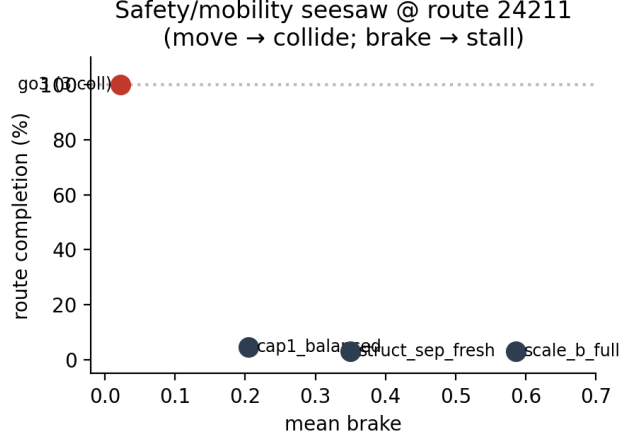


Figure 2: Every candidate sits at an extreme of the brake axis: high brake $\Rightarrow$ stall (route $\sim$ 3–5%), low brake $\Rightarrow$ collisions. The middle (selective deceleration) is never reached.

5.1 From escaping the over-brake to driving—and where it stops

Step 1: the over-brake was the learned brake. We change the one thing that produced it: the action head. The re-trained model (*identical* data, router, experts, gate, and route inputs) predicts steer and a forward command but **no brake**—the brake output is fixed to zero, so the model cannot over-brake; braking is delegated to a standard safety shield. This alone is a controlled change—only the action head’s brake differs from the structural fork—and it lifts the three gate routes from $\sim$ 3–4% (Table 3) to $\sim$ 46% completion, collision-free on two of three, with the agent never stalling.

Step 2: a control stack drives—and generalizes to some unseen routes. Pairing the learned target speed with a classical longitudinal controller (a PID), the safety shield, and a small anti-stall floor lets the policy actually drive. Of the five evaluation routes only 24211 appears in the training data; the other four are *unseen*. On an unseen route the policy drives cleanly end to end (Table 4): **route 24330 reaches 100% completion with zero collisions and zero lane departures**, and unseen 24240 reaches 53% (there it is permanently blocked behind a scenario obstacle that would require a lateral maneuver to pass).

Step 3: but it does not generalize *uniformly*. Two other unseen routes reach only 18–25% (Table 4), with collisions and lane departures—and even the lone *trained* route, 24211, still collides at 88%. So this is not a clean train/test overfitting gap (the cleanest route is unseen; the trained route is not perfect) but a *scenario-dependent* generalization gap: the policy handles some unseen scenarios cleanly and mishandles specific hazards in others, as expected for $\sim$ 8.5k frames over 43 routes with no on-policy data.

The honest reading is neither “it drives” nor “it fails.” With the control stack, the learned target-speed policy drives an *unseen* route cleanly to 100%—a first for this model—yet mishandles specific hazards on other unseen routes. The over-brake was the learned brake output; the remaining wall is *generalization to harder scenarios*, not the decomposition, and not a simple train/test overfit (the cleanest route is unseen; the one trained route still collides). (Completion comes from the policy *plus* the standard control stack, not the policy alone; results are single-checkpoint.)

Robustness, and what does *not* scale. The 24330 result is not a lucky single run: it holds at **100%**

Table 4: X-MoD with the target-speed head + a classical control stack (PID, safety shield, anti-stall floor). Cells are route completion % (collision count). Of the five routes only 24211 is in training; the other four are *unseen*. The policy drives one unseen route cleanly (24330, 100%/0, robust across three traffic seeds) but mishandles specific hazards on others—a scenario-dependent gap, not a clean train/test split (the trained route itself still collides).

	trained	unseen			
	24211	24330	24240	24781	24841
completion % (collisions)	88 (1)	100 (0)	53 (1)	25 (1)	18 (2)

completion with zero collisions across three traffic-manager seeds. The per-route *variation*, by contrast, tracks *training composition* rather than eval noise. Re-training on 11 of the 43 routes and re-evaluating the same unseen routes swings them widely (24330 $52 \rightarrow 100$, 24781 $81 \rightarrow 25$ as the training set grows $11 \rightarrow 43$), yet the *mean* unseen completion is essentially unchanged ($\approx 51\%$ at 11 routes vs. $\approx 49\%$ at 43). Quadrupling the training routes did not raise average generalization—it only redistributed which unseen routes the brittle policy happens to transfer to. Nor does adding *failure-correction* data help: re-training with the available privileged-expert collision-source corrections (the kind of data an on-policy DAgger loop would supply) *lowers* held-out completion (mean $\approx 22\% \rightarrow \approx 12\%$ with the failure states class-balanced, $\approx 8\%$ without) and degrades steering, the safety-heavy failure states reintroducing the very over-caution we had removed. So the residual is not a data-*quantity* wall, and it is not closed by the failure-correction data at hand; it is a composition/robustness problem that a small, single-camera policy does not absorb, pointing to capacity and a *carefully retained* on-policy loop rather than a quick data patch.

6 Why: Readable Arbitration $\neq$ Drivable Arbitration

The arbitration signal that makes X-MoD *explainable* (the gate) reads risk well, as Part I shows. But reading risk is not the same as *resolving* the safety $\leftrightarrow$ mobility conflict moment-to-moment, which is what control requires. The negative result localizes precisely what is missing:

1. **A closed-loop learning loop.** Nothing updates the policy’s priorities from the *consequences* of its own closed-loop states; imitation on clean teacher data never sees the student’s failure states (the classic compounding-error / distribution-shift wall).
2. **Persistence / memory.** A conditional response to an unfolding hazard needs state that survives across timesteps; a single feed-forward pass does not carry it.
3. **A monitor / “pause.”** No layer watches the arbitration and *interrupts* a bad trajectory; the gate explains but cannot veto.

Crucially, a fourth factor turned out to be **dominant**: the action head’s **brake output**. The four candidates all emit a learned brake; zeroing it (Table 4, with braking delegated to a standard safety shield) removes the over-brake stall on the identical model and data. So the over-brake seesaw was substantially an artifact of the learned brake output, *not* of interpretable decomposition. The three pieces above (loop, memory, monitor)—together with on-policy data—are what the *residual* failure now demands: with a control stack the policy drives some *unseen* routes cleanly (one to 100%) but mishandles specific hazards on others, and that scenario-dependent gap is exactly the distribution-shift the learning loop is meant to close. They are no longer needed to explain the over-brake corner, which the brake output alone accounts for.

7 Limitations and Non-claims

Claimed: (i) under *direct pedal regression*, an interpretable Mixture-of-Drives exhibits a reproducible over-brake/collision seesaw that survives data scale, loss balancing, width, and structural separation; (ii) removing the model’s learned brake output (an action head with no brake) and pairing the learned target speed with

a classical control stack (PID, safety shield, anti-stall floor)—changing neither the decomposition nor the data—lets the policy *complete* an *unseen* route (24330: 100%, zero collisions, *across three traffic seeds*; only 24211 of the five eval routes is in training). **Not claimed:** that this model *drives* Bench2Drive in general—it does *not* generalize uniformly: other unseen routes reach only 18–25% with collisions (and the lone trained route still collides at 88%), a scenario-dependent generalization gap rather than a clean train/test overfit; nor that the decomposition itself *helps* driving; nor that the gap is closed by *quantity*—a scaling probe (11 → 43 training routes) left mean unseen completion unchanged ($\approx 51\% \rightarrow \approx 49\%$). The Part I explanation result is primarily offline; runtime inputs are non-privileged; **completion comes from the learned policy *plus* a standard control stack** (PID + safety/anti-stall shields), not the policy alone; results are single-checkpoint (the 24330 milestone is three-seed).

8 Future Work—Toward Drivable Arbitration

The results set a sharp coordinate. The over-brake corner is solved by removing the learned brake, and the policy already drives an *unseen* route cleanly to 100%. The residual wall is **scenario-dependent generalization**: other unseen routes reach only 18–25%, the signature of pure imitation on $\sim 8.5\text{k}$ frames over 43 routes. Closing it is the harder road—on-policy data (Dagger over the policy’s own closed-loop states), more teacher data, and a lateral re-parametrization (waypoints + a classical lateral controller) for the steering errors that dominate the failing routes. The longer arc—a closed-loop learning loop, persistence, and a monitoring “pause”—is precisely what turns a trustworthy arbitration *signal* into a *generalizing* controller. The claims of *this* paper remain restricted to the measured results above.

9 Conclusion

“Readable” and “drivable” are not the same property—and for this interpretable policy, the gap between them was mostly its *learned brake output*, not the decomposition. Remove that brake and add a classical control stack, and the transparent arbitrator that explains risk faithfully (correlation 0.80) actually *drives*: it completes an *unseen* route at 100% with zero collisions. But it does not generalize uniformly—other unseen routes reach only 18–25%, and even the one trained route still collides. The honest verdict is neither “it fails” nor “it drives” but a located one: the decomposition was never the blocker; the over-brake was the learned brake, and what now stands between this policy and Bench2Drive is *generalization to harder scenarios*.

References

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